PREDICTIVE ANALYTICS PROBLEM SET 2 LINEAR REGRESSION

SULAGNA ROY

2026-01-31

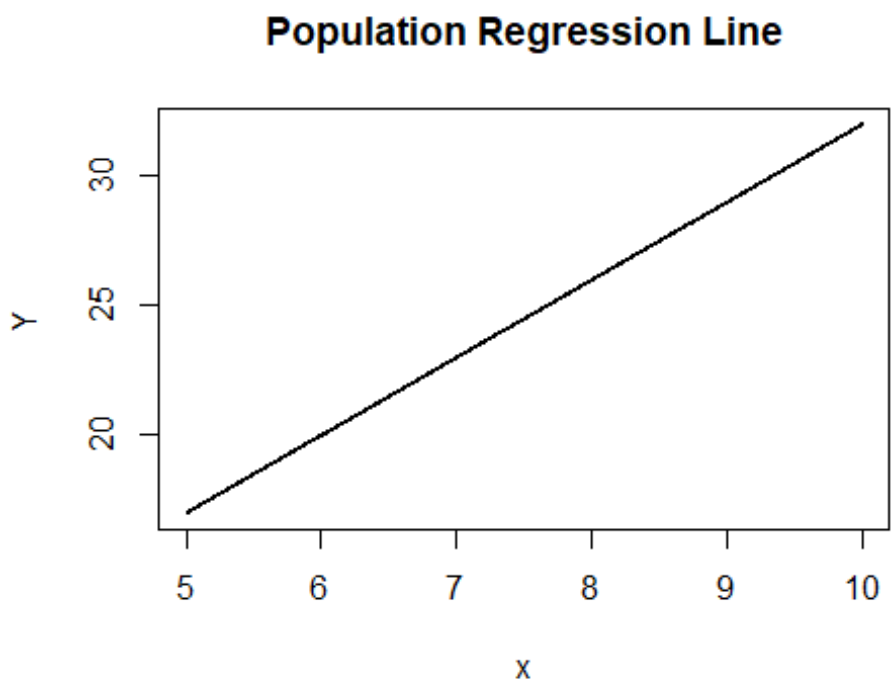
QUESTION 1.

Problem to demonstrate that the population regression line is fixed, but least square regression line varies

Suppose the population regression line is given by Y = 2 + 3x, while the data comes from the model y = 2+3x+E. (E:Random Error)

Step 1:For x in the range [5,10] graph the population regression line.

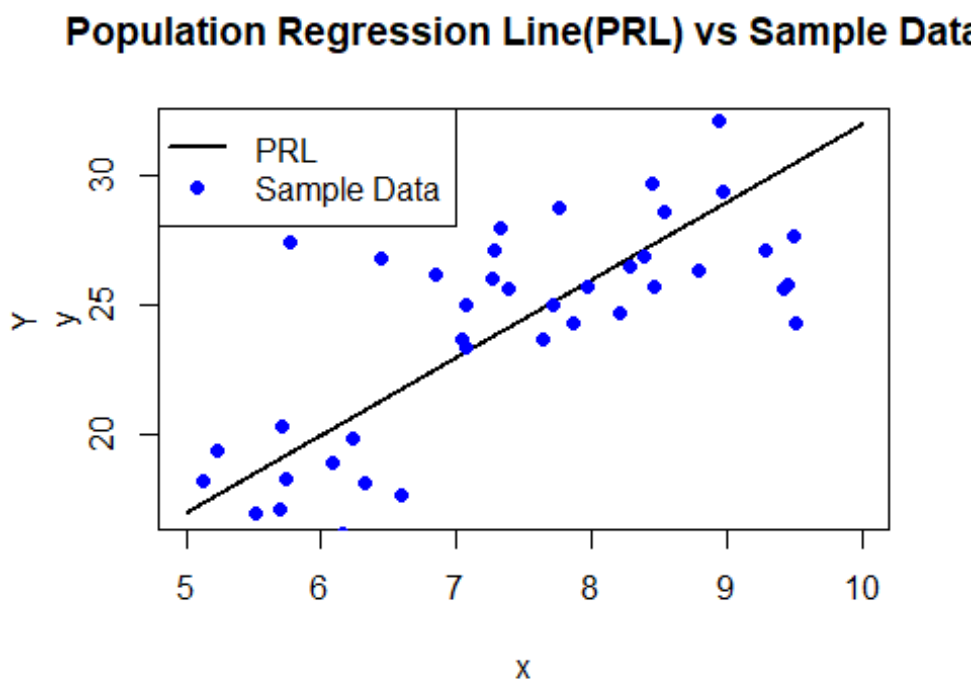
```
x=seq(5,10,length.out=1000)
Y=2+(3*x)
plot(x,Y,type="l",lwd=2,main="Population Regression Line")
```



Step 2:Generate xi(i = 1,2,..,n) from Uniform(5,10) and ei(i = 1,2,..,n) from N(0,4^2). Hence, compute y1,y2,..,yn.(later mentioned n=50, seed=123)

```
n=50
set.seed(123)
xi=runif(n,5,10)
ei=rnorm(n,0,4)
yi=2+(3*xi)+ei
plot(x,Y,ylab=c('Y','y'),type='l',lwd=2,main="Population Regression Line(PRL) vs
Sample Data")
points(xi,yi,pch=16,col='blue')
```

```
legend('topleft',
        legend=c('PRL',
                 'Sample Data'),
       col=c("black","blue"),
       pch = c(NA, 16),
       lty = c(1, NA),
       lwd = c(2, NA))
```

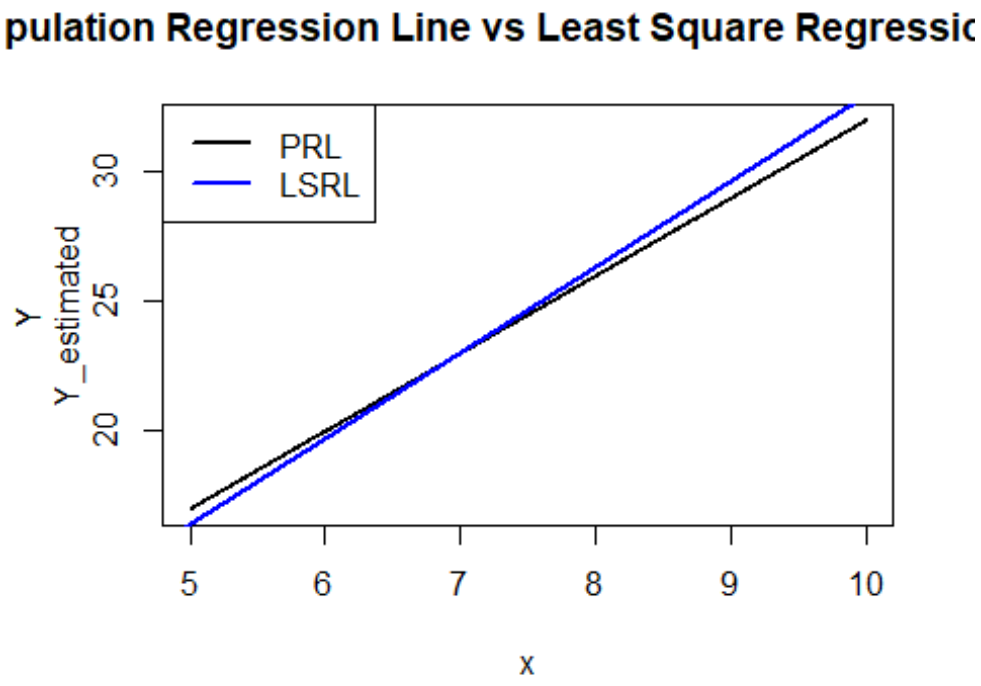


Step 3: On the basis of the data (xi,yi)(i = 1,2,..,n) generated in Step 2, report the least squares regression line.

```
n=50 #Sample Size
set.seed(123) #Seed value
xi=runif(n,5,10) #Fixed observations on predictor x
ei=rnorm(n,0,4) #Random error components
yi=2+(3*xi)+ei #Observed values on response
plot(x,Y,type='l',lwd=2) #Population Regression Line
lin_reg=lm(yi~xi) #Fitted Simple Regression Line obtained from sample data
coef(lin_reg) #Estimated values of model parameters

## (Intercept)          xi
## -0.09638929  3.30539569

abline(lin_reg,col='blue',lwd=2) #Fitted Simple Regression Line
legend('topleft',
        legend=c('PRL',
                 'LSRL'),
       col=c("black","blue"),
       lty=1,
       lwd=2)
title(main="Population Regression Line vs Least Square Regression Line",
      ylab=c('Y','Y_estimated'))
```

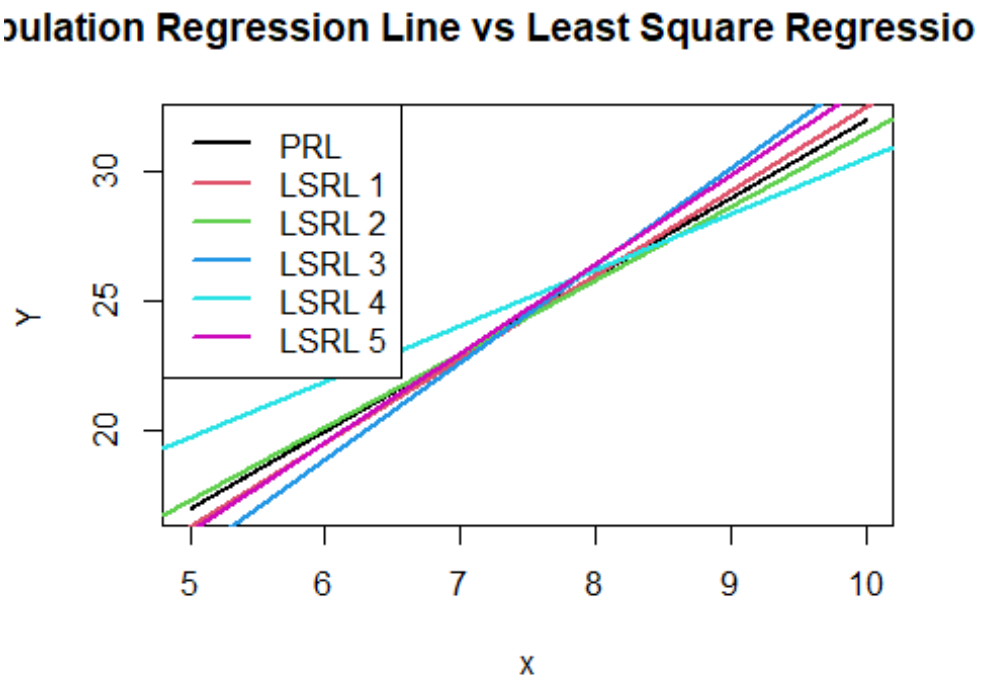


Step 4:Repeat steps 2-3 five times. Graph the 5 least squares regression lines over the population regression line obtained in Step 1. Interpret the findings.

```
set.seed(123)
xr=matrix(runif(n*5,5,10),ncol=5,nrow=n)
er=matrix(rnorm(n*5,0,4),ncol=5,nrow=n)
yr=matrix(0,ncol=5,nrow=n)
intercept=rep(0,5)
slope=rep(0,5)
plot(x,Y,type='l',lwd=2)
for(i in 1:5)
{
  yr[,i]=2+(3*xr[,i])+er[,i]
  lin_reg=lm(yr[,i]~xr[,i])
  intercept[i]=coef(lin_reg)[1]
  slope[i]=coef(lin_reg)[2]
  abline(lin_reg,col=i+1,lwd=2)
}
Estimates=data.frame(intercept,slope);Estimates

##    intercept    slope
## 1  0.1547357 3.238639
## 2  3.1053005 2.842175
## 3 -3.6350484 3.756428
## 4  9.0589133 2.147608
## 5 -1.0229227 3.432290

legend('topleft',
       legend=c("PRL",
                "LSRL 1",
                "LSRL 2",
                "LSRL 3",
                "LSRL 4",
                "LSRL 5"),
       col=1:6,
       lwd=2)
title(main="Population Regression Line vs Least Square Regression Lines")
```



Interpretation-

The Population Regression Line remains fixed where as the Least Square Regression Lines change with changing sample though they all closely estimate the Population Regression Line.

QUESTION 2.

Problem to demonstrate that beta_0_hat and beta_hat minimises RSS

Step 1: Generate xi from Uniform(5, 10) and mean centre the values. Generate ei from N(0,1). Calculate yi = 2 + 3xi + ei, i = 1,2,.., n. Take n=50 and seed=123.

```
n=50
set.seed(123)
x=runif(n,5,10)
z=x-mean(x)
set.seed(123)
e=rnorm(n)
y=2+3*z+e
```

Step 2: Now imagine that you only have the data on (xi,yi),i = 1,2,..,n, without knowing the mechanism that was used to generate the data in step 1. Assuming a linear regression of the type yi = beta_0 +beta*xi +ei, and based on these data (xi,yi),i = 1,2,..,n, obtain the least squares estimates of beta_0 and beta.

```
lin_reg=lm(y~z) #Fitted Model

#Estimates of parameters
beta_0_hat=coef(lin_reg)[1];beta_0_hat

## (Intercept)
##    2.034404

beta_hat=coef(lin_reg)[2];beta_hat
```

```
##         z
## 2.977528
```

Step 3: Take a large number of grid values of (beta_0,beta) that also include the least squares estimates obtained from step 2. Compute the RSS for each parametric choice of (beta_0,beta), where RSS = (y1 −beta_0 −beta*x1)^2 +(y2 −beta_0 −beta*x2)^2 +….(yn −beta_0 −beta*xn)^2. Find out for which combination of (beta_0,beta), RSS is minimum.

```
beta0.grid=seq(-1,3,length.out=101)+beta_0_hat
beta.grid=seq(-1,3,length.out=101)+beta_hat
#Computing RSS
RSS=c()
for(i in 1:length(beta.grid))
{
  RSS[i]=sum((y-beta0.grid[i]-beta.grid[i]*z)^2)
}
RSS

##   [1]  198.04277  185.80518  174.06707  162.82846  152.08934  141.84972
##   [7]  132.10959  122.86895  114.12781  105.88616   98.14401   90.90135
##  [13]   84.15818   77.91451   72.17033   66.92564   62.18045   57.93475
##  [19]   54.18855   50.94184   48.19462   45.94690   44.19867   42.94994
##  [25]   42.20070   41.95095   42.20070   42.94994   44.19867   45.94690
##  [31]   48.19462   50.94184   54.18855   57.93475   62.18045   66.92564
##  [37]   72.17033   77.91451   84.15818   90.90135   98.14401  105.88616
##  [43]  114.12781  122.86895  132.10959  141.84972  152.08934  162.82846
##  [49]  174.06707  185.80518  198.04277  210.77987  224.01645  237.75253
##  [55]  251.98811  266.72318  281.95774  297.69180  313.92535  330.65839
##  [61]  347.89093  365.62296  383.85448  402.58550  421.81602  441.54602
##  [67]  461.77552  482.50452  503.73301  525.46099  547.68846  570.41543
##  [73]  593.64190  617.36785  641.59331  666.31825  691.54269  717.26662
##  [79]  743.49005  770.21297  797.43538  825.15729  853.37869  882.09959
##  [85]  911.31998  941.03986  971.25924 1001.97811 1033.19648 1064.91434
##  [91] 1097.13169 1129.84854 1163.06488 1196.78071 1230.99604 1265.71086
##  [97] 1300.92518 1336.63899 1372.85229 1409.56509 1446.77738

df=data.frame(beta0.grid,beta.grid,RSS)
df

##     beta0.grid beta.grid        RSS
## 1     1.034404  1.977528  198.04277
## 2     1.074404  2.017528  185.80518
## 3     1.114404  2.057528  174.06707
## 4     1.154404  2.097528  162.82846
## 5     1.194404  2.137528  152.08934
## 6     1.234404  2.177528  141.84972
## 7     1.274404  2.217528  132.10959
## 8     1.314404  2.257528  122.86895
## 9     1.354404  2.297528  114.12781
## 10    1.394404  2.337528  105.88616
## 11    1.434404  2.377528   98.14401
## 12    1.474404  2.417528   90.90135
## 13    1.514404  2.457528   84.15818
## 14    1.554404  2.497528   77.91451
## 15    1.594404  2.537528   72.17033
## 16    1.634404  2.577528   66.92564
## 17    1.674404  2.617528   62.18045
## 18    1.714404  2.657528   57.93475
## 19    1.754404  2.697528   54.18855
## 20    1.794404  2.737528   50.94184
```

```
## 21    1.834404  2.777528   48.19462
## 22    1.874404  2.817528   45.94690
## 23    1.914404  2.857528   44.19867
## 24    1.954404  2.897528   42.94994
## 25    1.994404  2.937528   42.20070
## 26    2.034404  2.977528   41.95095
## 27    2.074404  3.017528   42.20070
## 28    2.114404  3.057528   42.94994
## 29    2.154404  3.097528   44.19867
## 30    2.194404  3.137528   45.94690
## 31    2.234404  3.177528   48.19462
## 32    2.274404  3.217528   50.94184
## 33    2.314404  3.257528   54.18855
## 34    2.354404  3.297528   57.93475
## 35    2.394404  3.337528   62.18045
## 36    2.434404  3.377528   66.92564
## 37    2.474404  3.417528   72.17033
## 38    2.514404  3.457528   77.91451
## 39    2.554404  3.497528   84.15818
## 40    2.594404  3.537528   90.90135
## 41    2.634404  3.577528   98.14401
## 42    2.674404  3.617528  105.88616
## 43    2.714404  3.657528  114.12781
## 44    2.754404  3.697528  122.86895
## 45    2.794404  3.737528  132.10959
## 46    2.834404  3.777528  141.84972
## 47    2.874404  3.817528  152.08934
## 48    2.914404  3.857528  162.82846
## 49    2.954404  3.897528  174.06707
## 50    2.994404  3.937528  185.80518
## 51    3.034404  3.977528  198.04277
## 52    3.074404  4.017528  210.77987
## 53    3.114404  4.057528  224.01645
## 54    3.154404  4.097528  237.75253
## 55    3.194404  4.137528  251.98811
## 56    3.234404  4.177528  266.72318
## 57    3.274404  4.217528  281.95774
## 58    3.314404  4.257528  297.69180
## 59    3.354404  4.297528  313.92535
## 60    3.394404  4.337528  330.65839
## 61    3.434404  4.377528  347.89093
## 62    3.474404  4.417528  365.62296
## 63    3.514404  4.457528  383.85448
## 64    3.554404  4.497528  402.58550
## 65    3.594404  4.537528  421.81602
## 66    3.634404  4.577528  441.54602
## 67    3.674404  4.617528  461.77552
## 68    3.714404  4.657528  482.50452
## 69    3.754404  4.697528  503.73301
## 70    3.794404  4.737528  525.46099
## 71    3.834404  4.777528  547.68846
## 72    3.874404  4.817528  570.41543
## 73    3.914404  4.857528  593.64190
## 74    3.954404  4.897528  617.36785
## 75    3.994404  4.937528  641.59331
## 76    4.034404  4.977528  666.31825
## 77    4.074404  5.017528  691.54269
## 78    4.114404  5.057528  717.26662
## 79    4.154404  5.097528  743.49005
```

```
## 80    4.194404  5.137528  770.21297
## 81    4.234404  5.177528  797.43538
## 82    4.274404  5.217528  825.15729
## 83    4.314404  5.257528  853.37869
## 84    4.354404  5.297528  882.09959
## 85    4.394404  5.337528  911.31998
## 86    4.434404  5.377528  941.03986
## 87    4.474404  5.417528  971.25924
## 88    4.514404  5.457528 1001.97811
## 89    4.554404  5.497528 1033.19648
## 90    4.594404  5.537528 1064.91434
## 91    4.634404  5.577528 1097.13169
## 92    4.674404  5.617528 1129.84854
## 93    4.714404  5.657528 1163.06488
## 94    4.754404  5.697528 1196.78071
## 95    4.794404  5.737528 1230.99604
## 96    4.834404  5.777528 1265.71086
## 97    4.874404  5.817528 1300.92518
## 98    4.914404  5.857528 1336.63899
## 99    4.954404  5.897528 1372.85229
## 100   4.994404  5.937528 1409.56509
## 101   5.034404  5.977528 1446.77738
```

```
df[which.min(RSS),]
```

```
##    beta0.grid beta.grid      RSS
## 26   2.034404  2.977528 41.95095
```

Comment: The Residual Sum of Squares is least when we use the Least Square Estimators of the model parameters beta_0 and beta.

QUESTION 3.

Problem to demonstrate that least square estimators are unbiased

Step 1: Generate xi(i = 1,2,..,n) from Uniform(0,1), ei(i = 1,2,..,n) from N(0,1) and hence generate y using yi = beta_0 +beta*xi +ei. (Take beta_0 = 2, beta = 3).

```
rm(list=ls())
n=50
set.seed(123)
x=runif(n)
e=rnorm(n)
beta_0=2
beta=3
y=beta_0+(beta*x)+e
```

Step 2:On the basis of the data (xi,yi)(i = 1,2,..,n) generated in Step 1, obtain the least square estimates of beta_0 and beta. Repeat Steps 1-2, R = 1000 times. In each simulation obtain beta_0_est and beta_est. Finally, the least-square estimates will be given by the average of these estimated values. Compare these with the true beta_0 and beta and comment. Take n = 50 and seed=123.

```
lin_reg=lm(y~x)
coef(lin_reg)
```

```
## (Intercept)           x
##    1.857647    3.381745
```

```
R=1000
set.seed(123)
```

```
xr=matrix(runif(1000*n),ncol=n,nrow=1000)
set.seed(123)
er=matrix(rnorm(1000*n),ncol=n,nrow=1000)
yr=matrix(0,ncol=n,nrow=1000)
beta_0_est=rep(0,1000)
beta_est=rep(0,1000)
for(i in 1:1000)
{
  for(j in 1:n)
  {
    yr[i,j]=beta_0+beta*xr[i,j]+er[i,j]
  }
  lin_reg=lm(yr[i,]~xr[i,])
  beta_0_est[i]=coef(lin_reg)[1]
  beta_est[i]=coef(lin_reg)[2]
}
#Mean of beta_0 estimates(An estimate of beta_0)
mean_beta_0_est=mean(beta_0_est);mean_beta_0_est

## [1] 1.990667

#Mean of beta estimates(An estimate of beta)
mean_beta_est=mean(beta_est);mean_beta_est

## [1] 3.013692

#Sd of beta_0 estimates(An estimate of Standard error of beta_0 estimator)
se_beta_0_est=sd(beta_0_est);se_beta_0_est

## [1] 0.2809015

#Sd of beta estimates(An estimate of Standard error of beta estimator)
se_beta_est=sd(beta_est);se_beta_est

## [1] 0.4894995
```

Interpretation-

Mean of Estimates:

The Arithmetic mean means of beta_0 and beta estimates converges to the true beta_0 and beta, thus Least Square Estimators are unbiased.

Variance of Estimates:

The variance of the estimates reduces as the no. of repeated sampling (R, here=1000) is increased, thus Least Square estimators are Consistent.

QUESTION 4:

Comparing several simple linear regressions

Attach “Boston” data from MASS library in R. Select median value of owner occupied homes, as the response and per capita crime rate, nitrogen oxides concentration, proportion of blacks and percentage of lower status of the population as predictors.

```
library(MASS)
attach(Boston)
y=medv
x1=crim
x2=nox
```

```
x3=black
x4=lstat
```

(a) Selecting the predictors one by one, run four separate linear regressions to the data. Present the output in a single table.

```
fit1=lm(y~x1)
fit2=lm(y~x2)
fit3=lm(y~x3)
fit4=lm(y~x4)
library(stargazer)

## Warning: package 'stargazer' was built under R version 4.5.2

##
## Please cite as:

##  Hlavac, Marek (2022). stargazer: Well-Formatted Regression and Summary Statistics
Tables.

##  R package version 5.2.3. https://CRAN.R-project.org/package=stargazer

stargazer(fit1,fit2,fit3,fit4,type="text",out="f.txt")

##
## ===========================================================================
##                                           Dependent variable:
##                                   ----------------------------------------
##                                                    y
##                                      (1)       (2)        (3)       (4)
## ---------------------------------------------------------------------------
## x1                                -0.415***
##                                    (0.044)
##
## x2                                          -33.916***
##                                              (3.196)
##
## x3                                                     0.034***
##                                                         (0.004)
##
## x4                                                               -0.950***
##                                                                   (0.039)
##
## Constant                          24.033*** 41.346***  10.551*** 34.554***
##                                    (0.409)   (1.811)    (1.557)   (0.563)
##
## ---------------------------------------------------------------------------
## Observations                        506       506        506       506
## R2                                 0.151     0.183      0.111     0.544
## Adjusted R2                        0.149     0.181      0.109     0.543
## Residual Std. Error (df = 504)     8.484     8.323      8.679     6.216
## F Statistic (df = 1; 504)        89.486*** 112.591*** 63.054*** 601.618***
## ===========================================================================
## Note:                                          *p<0.1; **p<0.05; ***p<0.01
```

(b) Which model gives the best fit?

Under the restriction that the 4 models have low p value, we say, the fitted model of our response, median value of house prices on the 4th predictor i.e., percent population of lower status of people gives the best fit because, it has the highest R^2 value, lowest Residual Standard error.

(c) Compare the coefficients of the predictors from each model and comment on the usefulness of the predictors.